

CHEN ZHIWEI

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SUMMARY

I am an AI research applicant interested in LLM agents for strategic reasoning and decision-making. My recent work focuses on building general LLM-based agents for imperfect-information strategic games, with prior experience in multimodal representation learning, image-text alignment, and visual question answering.

EDUCATION

National University of Singapore

Aug 2023 - Nov 2024

Master of Technology in Artificial Intelligence Systems, GPA: 3.51/4.0

Sun Yat-sen University

Sep 2018 - Jun 2022

Bachelor of Engineering in Traffic Engineering, GPA: 3.5/4.0

RESEARCH EXPERIENCE

Personality-Conditioned Lookahead for Evolving LLM Gaming Agents Apr 2026 - Jul 2026

Investigating whether LLM agents can adaptively evolve in repeated imperfect-information strategic gameplay against opponents that might be non-stationary, including Leduc Hold'em, Liar's Dice, and Goofspiel.

- Formulated online adaption for LLM gaming agents under non-replayable interactions as a key challenge for LLM game-playing agents.
- Proposed PLSE, a training-free decision-time adaptation framework that combines evolving opponent belief modeling, equilibrium-guided lookahead, and failure-aware reasoning.
- Evaluated on Leduc Hold'em, Liar's Dice, and Goofspiel with diverse opponent settings, showing improved performance over equilibrium-based baseline and conventional LLM-agent method.

Contrastive Ultrasound Report Generation

Apr 2025 - Aug 2025

Investigated medical report generation under more diverse imaging characteristics, moving beyond prior radiology-oriented methods that mainly optimized local cross-modal attention under relatively fixed imaging settings.

- Designed EchoDice, a balanced and diversity-aware sampling strategy that encourages fine-grained image-caption learning across different anatomical categories.
- Proposed ReportMatcher, a CLIP-style alignment module that jointly trains image caption task with image-text contrastive learning to improve global multimodal representation alignment.
- Achieved state-of-the-art performance on ultrasound report generation benchmarks, with strong transferability to radiology report generation using the same training framework.

WORKING EXPERIENCE

The Hong Kong University of Science and Technology, Guangzhou

Research Assistant, mentored by Prof T. Zhao

Apr 2026 - Present

- Conducted literature review on LLM based on-policy distillation.
- Developed experimental environments and evaluation pipelines for Leduc Hold'em, Liar's Dice, and Goofspiel.

Center for Artificial Intelligence and Robotics, Chinese Academy of Sciences, Hong Kong

Intern (24-25), Research Assistant (25-26), mentored by Prof M. Zhao

Mar 2024 - Feb 2026

- Developed CycleGAN-based thyroid reconstruction models for high-field imaging.
- Built a PDF data-cleaning pipeline for MLLM pretraining.
- Designed web crawlers for medical data acquisition and dataset construction.
- Evaluated the multimodal reasoning capabilities of large multimodal models on medical question-answering tasks.
- Conducted ultrasound multimodal research in image captioning and image-text alignment.

PUBLICATIONS

- Z. Rao, P. Zhou, D. Chong, **Z. Chen**, S. Wang, N. Tang. Structural and Disentangled Adaptation of Large Vision Language Models for Multimodal Recommendation. *SIGIR*, 2026

UNDER REVIEW

- **Z. Chen**, T. Wang, T. Zhao. Personality-Conditioned Lookahead for Evolving LLM Gaming Agents. Submitted to *AAAI*, 2027
- **Z. Chen**, Z. Rao, J. Chen, X. Li. Contrastive Ultrasound Report Generation with Diversity-Aware Learning Network, Submitted to *CIKM*, 2026